

Reference	Dataset	Methodology	Evaluation Metrics	Results	Gap Analysis
Karpakam, S., Kumareshan, 2025[]	• FigShare • MRI • X-ray	• Adaptive median-bilateral filtering • DSAS • GLCM • CDA features • DIR-GAN with attention-residual connections	• Accuracy • Precision • Recall • F1	98.90% FigShare	
98.60% MRI 97.93% X-ray	• Focus on synthetic image quality • Lacks cross-institutional clinical validation				
Zahid Ullah, Jihie Kim, 2025[]	• Two public Kaggle MRI sets	• Dual-level ensembling • Feature-level fusion of V16 classifier-level ensemble with optimized ML	• Accuracy • AUC reported	Outperforms SOTA (exact number not stated in abstract)	• Pre-print: lacks peer review • Clinical test-time deployment • Limited dataset diversity

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Muhammad Usman Akbar et al., 2024[7]	• BraTS 2020: 369 subjects • BraTS 2021: 1251 subjects • MR modalities: T1w, T1wCe, T2w, and FLAIR	• Progressive GAN, StyleGAN (G1–G3), and diffusion models • Synthetic MR image generation • U-Net and Swin Transformer for segmentation	• Dice Score • Hausdorff Distance • Fréchet Inception Distance (FID) • Inception Score (IS)	Synthetic images trained on GANs achieved Dice scores of 91%–100%	• Memorization of training data • Metric limitations • High computational cost • Impact of hyperparameters • Data variability
(Alshwan, Nasser et al., 2024)[P*]	• BrSHI Brain MRI: non-tumorous • Yes: 1500 tumorous images • No: 1500 non-tumorous images • Pred: 60 images for testing/prediction	• Synthetic Generative Adversarial Networks (GANs) • Federated Convolutional Neural Networks (CNNs)	• Sensitivity • Specificity • Precision • F1 Score	Accuracy of 99.82% achieved using DenseNet121 for feature extraction and classification.	• Low accuracy • Data scarcity • Privacy issues

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Wenxiu Wang et al., 2023[?]	• BraTS 2020: 369 patients • ICTS: 192 T1ce MR patients • In-house Dataset: 50 glioma patients	• Two-Stage Generative Model (TSGM) • CycleGAN + Variance Exploding Joint Probability (VE-JP) diffusion model	• Dice Similarity Coefficient (DSC) • Precision • Recall • Area Under the Precision-Recall Curve (AUPRC) • Intersection over Union (IoU) • Hausdorff95 Distance (HD95)	Achieved a SC of 85.9% on BraTS2020, surpassing traditional methods with accurate segmentation of complex tumor boundaries.	• Improve edge detection for complex anatomical regions • Optimize multi-modality data integration • Enhance computational efficiency for faster image generation • Better utilize multi-channel data to reduce interference • Expand validation across diverse datasets and clinical conditions

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Result: Outperformed human-designed and NAS-based methods	• Focuses only on MRI-based neural architecture optimization • Does not incorporate multi-modal data (e.g., medical history, lab results) • May limit classification accuracy and clinical applicability				